Appendix C

Item-to-Indicator Mapping for the PLS-SEM Measurement Model

Table B1 reports the complete mapping from questionnaire items, as administered to participants at the school and reproduced in full in Appendix A, to the indicator variables that enter the PLS-SEM measurement model. The second column references the question number in the Appendix A questionnaire; because the ERA-RSI block was administered in three parallel versions depending on the platform selected in question 11 (TikTok: questions 12–37; Instagram: questions 38–63; other platforms: questions 64–88), the three parallel question numbers are reported (TikTok / Instagram / other platforms). The fourth column references the corresponding item number in each source instrument: the Risk of Addiction–Adolescent to Social Networks and Internet Scale (ERA-RSI; Peris et al., 2018), the Pittsburgh Sleep Quality Index (PSQI; Buysse et al., 1989; Spanish adaptations: Moreno et al., 1996; Royuela et al., 2002), the Rosenberg Self-Esteem Scale (RSE; Rosenberg, 1965; Spanish adaptation: Atienza et al., 2000), and the Depression and Anxiety Interference Scale for Children and Adolescents (EIDAN; Espinosa et al., 2022). Items marked (R) in the source-scale column were reverse-coded prior to aggregation. Sociodemographic items (questions 1–2 and 5–10) are not listed, as they do not contribute to any indicator variable but instead function as descriptive or grouping variables in the multigroup analyses (MGA) (Section 4.3); the informed-consent items (questions 3–4) are likewise omitted. Also not listed is the platform-selection item (red_principal, question 11: “Which of these social media platforms do you use the most?”; 0 = I do not use social media, 1 = TikTok, 2 = Instagram, 3 = Facebook, 4 = YouTube, 5 = Other), which served as the questionnaire’s branching variable: participants selecting option 0 were routed to a parallel non-user branch (questions 117–146) and

excluded from the analytic sample (see Section 3.3), and the remaining participants received the ERA-RSI block with item wording tailored to their selected platform.

Table B1

Item-to-Indicator Mapping From the Questionnaire to the PLS-SEM Measurement Model

Indicator	Item (Appendix A)	Content (abbreviated, English)	Source-scale item
era_sint	13 / 39 / 64	Daily time spent on [social media platform] (1–4)	ERA-RSI item 1
era_sint	14 / 40 / 65	Use [social media platform] during study or work hours	ERA-RSI item 2
era_sint	15 / 41 / 66	Would feel angry if forced to give up [social media platform]	ERA-RSI item 3
era_sint	16 / 42 / 67	Access [social media platform] anywhere and anytime	ERA-RSI item 4
era_sint	17 / 43 / 68	[social media platform] has affected school grades	ERA-RSI item 5
era_sint	18 / 44 / 69	Lost sleep due to [social media platform] use	ERA-RSI item 6
era_sint	19 / 45 / 70	Hide [social media platform] usage time at home	ERA-RSI item 7
era_sint	20 / 46 / 71	Feel insecure without internet access	ERA-RSI item 8
era_sint	21 / 47 / 72	Update my status	ERA-RSI item 9
era_uso	22 / 48 / 73	Check friends' profiles	ERA-RSI item 10
era_uso	23 / 49 / 74	Use the chat function	ERA-RSI item 11
era_uso	24 / 50 / 75	Share photos and/or videos	ERA-RSI item 12
era_uso	25 / 51 / 76	Comment on photos with friends	ERA-RSI item 13
era_uso	26 / 52 / 77	See what contacts have been doing recently	ERA-RSI item 14
era_uso	27 / 53 / 78	Reconnect with former friends	ERA-RSI item 15
era_uso	28 / 54 / 79	Number of videos published (rescaled 0–3 → 1–4)	ERA-RSI item 16
era_uso	29 / 55 / 80	Make new friends	ERA-RSI item 17
era_fri	30 / 56 / 81	Play virtual and/or role-playing games	ERA-RSI item 18
era_fri	31 / 57 / 82	Join special-interest groups	ERA-RSI item 19
era_nomo	32 / 58 / 83	Feel safer being able to communicate anytime	ERA-RSI item 24
era_nomo	33 / 59 / 84	Safer to send a photo by phone than on social media	ERA-RSI item 25
era_nomo	34 / 60 / 85	Anxiety when messages are not answered immediately	ERA-RSI item 26
era_nomo	35 / 61 / 86	Would feel furious if mobile phone were taken away	ERA-RSI item 27
era_nomo	36 / 62 / 87	Need to know whether the recipient has read my message	ERA-RSI item 28
era_nomo	37 / 63 / 88	Uneasy when no one talks to me online	ERA-RSI item 29
psqi	89	Usual bedtime (last month)	PSQI item p1
psqi	90	Time to fall asleep, in minutes (last month)	PSQI item p2
psqi	91	Usual wake-up time (last month)	PSQI item p3
psqi	92	Actual hours of sleep per night (last month)	PSQI item p4
psqi	93	Trouble sleeping: cannot fall asleep within 30 min	PSQI item p5a
psqi	94	Trouble sleeping: wake during the night or early morning	PSQI item p5b
psqi	95	Trouble sleeping: need to get up to use the bathroom	PSQI item p5c
psqi	96	Trouble sleeping: cannot breathe comfortably	PSQI item p5d

Indicator	Item (Appendix A)	Content (abbreviated, English)	Source-scale item
psqi	97	Trouble sleeping: cough or snore loudly	PSQI item p5e
psqi	98	Trouble sleeping: feel too cold	PSQI item p5f
psqi	99	Trouble sleeping: feel too hot	PSQI item p5g
psqi	100	Trouble sleeping: had bad dreams or nightmares	PSQI item p5h
psqi	101	Trouble sleeping: had pain	PSQI item p5i
psqi	102	Trouble sleeping: other reasons (free text; excluded from C5)	PSQI item p5j
psqi	103	Overall subjective sleep quality (last month)	PSQI item p6
psqi	104	Use of sleep medication (last month)	PSQI item p7
psqi	105	Daytime sleepiness at school or other activities	PSQI item p8
psqi	106	Lack of motivation for daily activities	PSQI item p9
Self-Esteem	107	I feel I am a person of worth, equal to others	RSE item 1
Self-Esteem	108	I tend to feel that I am a failure (reverse-coded)	RSE item 2 (R)
Self-Esteem	109	I feel I have a number of good qualities	RSE item 3
Self-Esteem	110	I am able to do things as well as most other people	RSE item 4
Self-Esteem	111	I feel I do not have much to be proud of (reverse-coded)	RSE item 5 (R)
Self-Esteem	112	I take a positive attitude toward myself	RSE item 6
Self-Esteem	113	On the whole, I am satisfied with myself	RSE item 7
Self-Esteem	114	I wish I could have more respect for myself (reverse-coded)	RSE item 8 (R)
Self-Esteem	115	I certainly feel useless at times (reverse-coded)	RSE item 9 (R)
Self-Esteem	116	At times I think I am no good at all (reverse-coded)	RSE item 10 (R)
eidan_school	148	Difficulty doing homework or studying	EIDAN item 2 (F1)
eidan_school	150	Difficulty with daily activities (eating, sleeping, etc.)	EIDAN item 4 (F1)
eidan_school	152	Did not perform on exams as well as could have	EIDAN item 6 (F1)
eidan_school	155	Difficulty paying attention or asking questions in class	EIDAN item 9 (F1)
eidan_family	147	Difficulty getting along with siblings or other family	EIDAN item 1 (F2)
eidan_family	151	Difficulty communicating with parents	EIDAN item 5 (F2)
eidan_family	154	Difficulty doing pleasant activities with family	EIDAN item 8 (F2)
eidan_peers	149	Difficulty relating to classmates at school	EIDAN item 3 (F3)
eidan_peers	153	Difficulty or reluctance to meet up with friends	EIDAN item 7 (F3)
eidan_peers	156	Difficulty making new friends	EIDAN item 10 (F3)

Note. Indicator names match the variable labels used in the PLS-SEM specification (see Table 1). Question numbers correspond to the order of presentation in the institutional questionnaire reproduced in Appendix A; for ERA-RSI items, the three numbers refer to the TikTok, Instagram, and other-platforms branches, respectively. Source-scale item numbers refer to the canonical numbering in the respective validation studies. ERA-RSI = Risk of Addiction–Adolescent to Social Networks and Internet Scale (Peris et al., 2018); PSQI = Pittsburgh Sleep Quality Index (Buysse et al., 1989); RSE = Rosenberg Self-

Esteem Scale (Rosenberg, 1965); EIDAN = Depression and Anxiety Interference Scale for Children and Adolescents (Espinosa et al., 2022). (R) = reverse-coded prior to aggregation. F1, F2, and F3 denote the three factors of the EIDAN scale (school and daily activities, family, peers, respectively). [social media platform] denotes the participant's primary platform, as selected via the branching item red_principal, and item wording was tailored accordingly. Items 20–23 of the original ERA-RSI freaky-traits subscale (erotic-sexual content) were excluded a priori from the questionnaire on ethical grounds. The device item (questions 12 and 38; not administered in the other-platforms branch), which records the device used to access the social media platform, is contextual and does not enter the computation of any indicator variable; it is therefore not listed. Question 102 (PSQI p5j) was administered as free text and is therefore not included in the computation of PSQI component C5.

References

Atienza, F. L., Moreno, Y., & Balaguer, I. (2000). An analysis of the dimensionality of the Rosenberg self-esteem scale in a sample of Valencian adolescents. *Revista de Psicología*, *22*(12), 29-42.

Buysse, D. J., Reynolds III, C. F., Monk, T. H., Berman, S. R., & Kupfer, D. J. (1989). The Pittsburgh Sleep Quality Index: a new instrument for psychiatric practice and research. Psychiatry research, 28(2), 193-213.

Espinosa, V., Sandin, B., Garcia-Escalera, J., Valiente, R. M., & Chorot, P. (2022). Depression and Anxiety Interference Scale for Children and Adolescents: Factorial Structure and Psychometric Properties/Escala de Interferencia de la Depresión y la Ansiedad para Niños y Adolescentes (EIDAN): Estructura

Factorial y Propiedades Psicométricas. *Revista Iberoamericana de Diagnóstico y Evaluación-e Avaliação Psicológica*, (65), 19-34.

Moreno, P. L., Aguado, J. A., Royuela, A., & Macías, J. A. (1996). Influence of examinations in sleep quality of the students in medicine in Valladolid evaluated by PSQI. *European Psychiatry*, *11*(S4), 392s-392s.

Peris, M., Maganto, C., & Garaigordobil, M. (2018). Scale of risk of addiction to social networks and Internet for adolescents: reliability and validity (ERA-RSI). Revista de Psicología Clínica con Niños y Adolescentes, 5(2), 30-36.

Rosenberg, M. (1965). *Society and the adolescent self-image* (Vol. 11, p. 326). Princeton, NJ: Princeton university press.

Royuela, A., Macias, J. A., Gil-Verona, J. A., Pastor, J. F., Maniega, M. A., Alonso, J., ... & Boget, T. (2002). Sleep in schizophrenia: a preliminary study using the Pittsburgh Sleep Quality Index. *Neurobiology of Sleep-Wakefulness Cycle*, *2*(2), 37-39.